

Tuberculosis – Summary

Legend:

- | | | |
|--------------------|----------------|------------------------------|
| ● EMB: Ethambutol | INH: Isoniazid | PZA: Pyrazinamide |
| ● RIF: Rifapentine | RMP: Rifampin | DOT: Direct Observed Therapy |

Key Symptoms: Weight loss, fatigue, fever, night sweats, hemoptysis, pleural pain, hoarseness, cough lasting > 2 weeks with/without blood and/or phlegm, dyspnea, etc.

Most common causing-pathogens: *Mycobacterium tuberculosis*

Diagnosis: Risk factors, chest x-ray, blood panel, sputum culture, liver enzymes, Mantoux skin test, etc.

Non-Pharmacological Treatment: Isolation at home, appropriate PPE e.g. N95 masks, avoid alcohol and other hepatotoxic agents, promote smoking cessation, TDM, optimal co-morbidity management

Pharmacological Treatment:

- **Latent:** RMP daily x 4 months **OR** INH + RIF weekly x 3 months
 - Rifapentine is not approved in Canada
- **2nd line:** INH daily x 9 months
- **Active**
 - **Intensive phase:** INH + RMP + PZA +/- EMB daily x 2 months **OR** INH + RMP + EMB daily x 2 months (for patients > 75 yrs old)
 - **Continuation phase:** INH + RMP daily or 3x/week for 4-7 months
- **Pregnancy:** INH + RIF + EMB is recommended as initial therapy
 - PZA is unlikely to be teratogenic
- **Breastfeeding:** is safe with INH, RMP, PZA, EMB + pyridoxine

Monitoring Parameters: Clinical monitoring (x-rays, symptoms, etc.), smears & cultures, drug-specific monitoring and routine lab monitoring (LFTs, CBC, hep B/HIV serology, etc.)

Additional Key Points:

- The Mantoux/tuberculin skin test may yield false positives in those who have received the BCG vaccine and may yield false negatives in the elderly, immunocompromised, severe illness
- Those with a latent TB infection have a 5-10% lifetime chance of developing active TB
 - Risk increases in those who are immunocompromised, HIV+, elderly, etc.
- Always add pyridoxine (Vit B6) with isoniazid because of peripheral neuropathy S/E risk (reversible upon cessation)
- Risk for drug-induced hepatitis: PZA > INH > RIF > EMB
- Baseline and monthly visual acuity testing required with EMB due to optic neuritis risk
- HIV + patients should be treated with rifabutin > RMP due to drug interactions between RMP and many antiretrovirals
- DOT is especially recommended in high-risk populations e.g. hx of tx failure/relapse, HIV + patients, individuals who experience substance abuse disorders and/or mental health disorders, etc.